Corrected neural-lexical model ensemble: Methodology and Evidence

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Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Combine complementary lexical and neural retrieval evidence while enforcing the seven-eligible-case contract. This is a prediction ensemble built from separately trained branches, not a single retriever architecture.

Architecture and mechanism

The selected full/partial neural scores and lexical TF-IDF scores are row-normalized and blended. The corrected export removes self, fills any missing seventh slot with eligible candidates, and validates uniqueness and pool membership.

Measured result

Scored submission	1039898, 1039902
Codabench micro-F1@7	0.4064
Archived local metric	not independently reported after correction

The corrected ensemble scored 0.4064 Codabench micro-F1@7, tied to four decimals with the corrected full retriever. Its earlier 0.4158 local dev figure predates the self-match fix and must not be presented as corrected dev.

Limitations and next check

The correction report leaves corrected ensemble dev metric null; do not substitute the earlier optimistic figure. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: phase7_export.py, phase7_run.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.